Exercise 1

Given is the following task set.

process i	phase Φ_i	computation time C_i	period T_i	deadline D_i
τ_1	0	2	5	5
τ_2	0	4	7	7

→ RT Analysis?

a) Can schedulability be guaranteed under RM or EDF?

b) Try to find a schedule with EDF and RM.

Theorem:
A set of periodic tasks $\tau_1, \ldots, \tau_n$ (with $D_i = T_i$) is schedulable with EDF iff $\sum_{i=1}^{n} C_i / T_i \leq 1$.

Any set of n periodic tasks with a processor utilization factor $\leq \boxed{U_{lub} = n(2^{1/n} - 1)}$ can be scheduled by RM.

Any set of periodic tasks with a processor utilization factor $\leq$ ln 2 can be scheduled by RM.

This is a sufficient condition, but not a necessary condition!

a) $U = \frac{2}{5} + \frac{4}{7} = \frac{14+20}{35}$ EDF ✓

$RM^{\times} = \frac{34}{35} \leq 1$

$\not\leq 2 \cdot (\sqrt[2]{2} - 1)$

$\approx 0{,}83$

$\frac{1}{5}$ $\frac{1}{7}$

prio(τ_1) > prio(τ_2) (smaller period first)

Because not necessarily unsched. with RM → Response Time An.

$$R_i^{(0)} = C_i \leq D_i$$

$$R_i^{(j+1)} = C_i + \sum_{k=1}^{i-1} \left\lceil \frac{R_i^{(j)}}{T_k} \right\rceil \cdot C_k \leq D_i$$

$$R_1^{(0)} = 2 \leq 5 = D_1$$

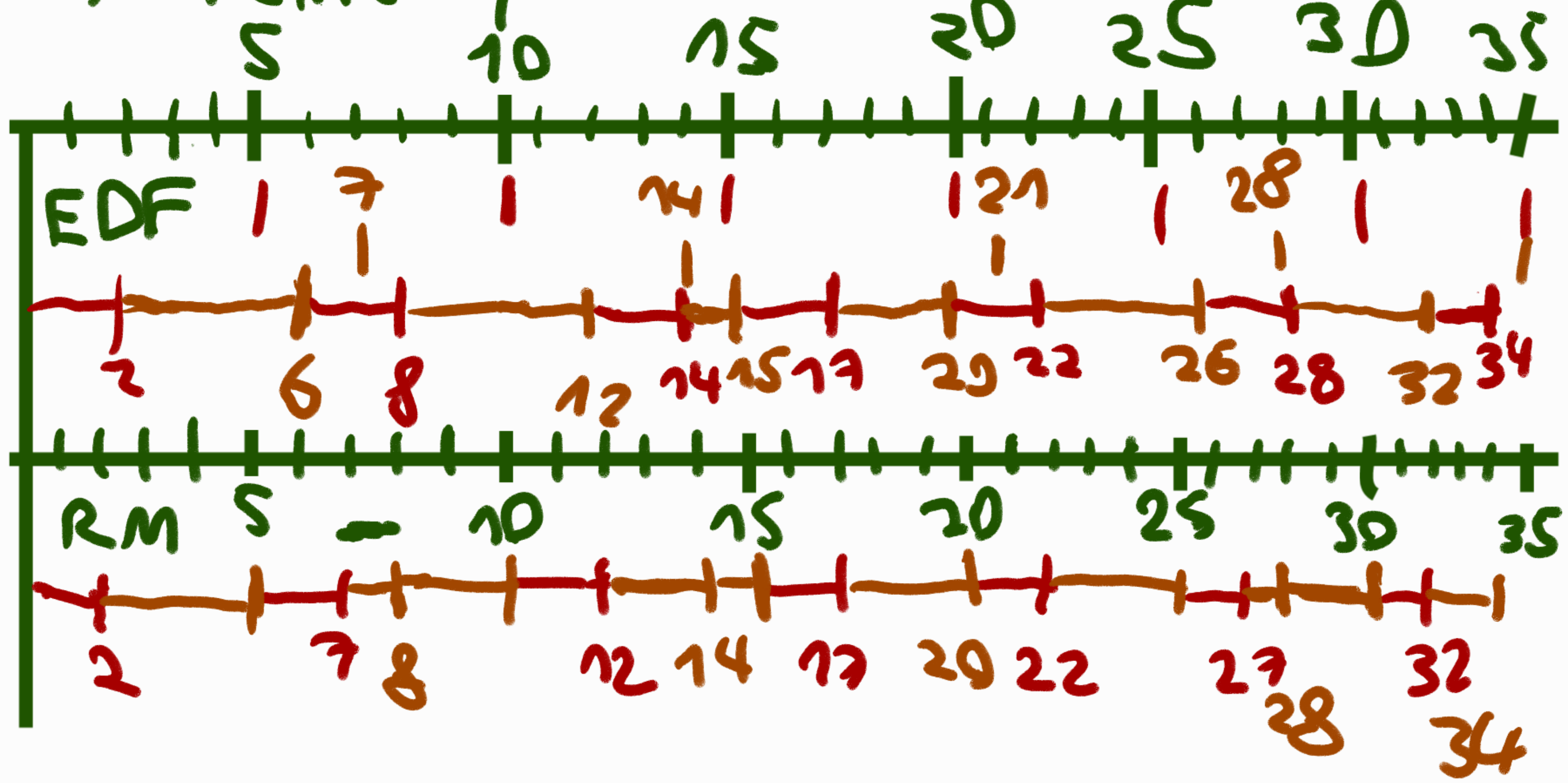
$$R_2^{(0)} = 4 \leq 7 = D_2$$

$$R_2^{(1)} = 4 + \left\lceil \frac{4}{5} \right\rceil \cdot 2 = 6 \leq 7$$

$$= 4 + \left\lceil \frac{6}{5} \right\rceil \cdot 2 = 8 \not\leq 7$$

⇝ not schedulable

b) lcm(5, 7) = 35



Exercise 3

Given is the following task set of periodic tasks

process i	phase Φ_i	computation time C_i	period T_i	deadline D_i
τ_1	0	3	10	10
τ_2	0	2	4	4

Also given is the following task set of aperiodic tasks

process i	arrival time a_i	computation time C_i
τ_I	2	1
τ_{II}	5	1
τ_{III}	9	1

↳ own priority: $prio(\tau_I) > prio(\tau_{II}) > prio(\tau_{III})$

a) Give a schedule when the periodic tasks are scheduled using EDF and the aperiodic tasks are scheduled in the background.

↝ doesn't work for RM ?

b) Compute the values of U_p and U_s according to the lecture.

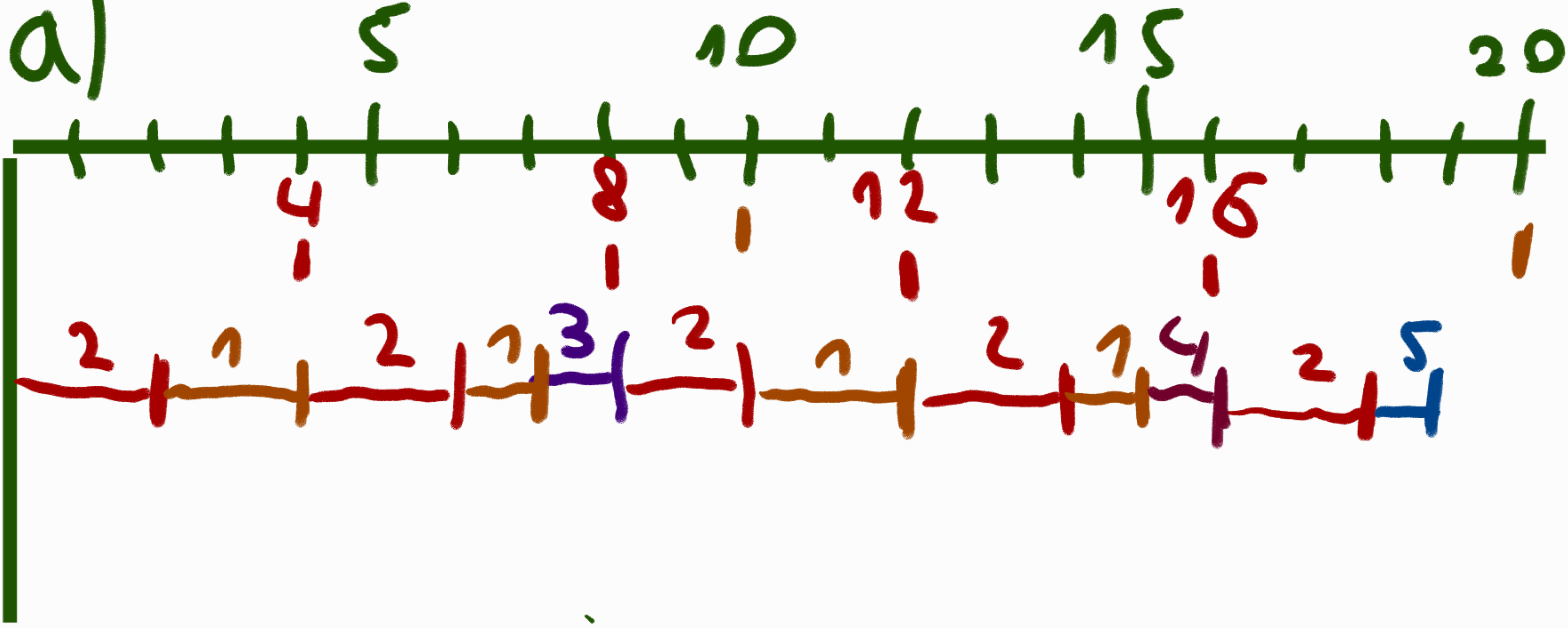
?

c) Compute the deadlines of the aperiodic tasks using the Total Bandwidth Server (TBS) algorithm.

d) Schedule both the periodic and aperiodic tasks with deadlines computed according to TBS together using the EDF algorithm.

lcm(10, 4) = 20

a)



b)

$$u_p = \frac{3}{10} + \frac{2}{4} = \frac{6+10}{20} = \frac{16}{20} \overset{:5}{=} \frac{4}{5}$$

$$u_s = 1 - \frac{4}{5} = \frac{1}{5}$$

↳ sparse

c)

$$d_{r_I} = 2 + \frac{1}{\frac{1}{5}} = 2 + 5 = 7$$

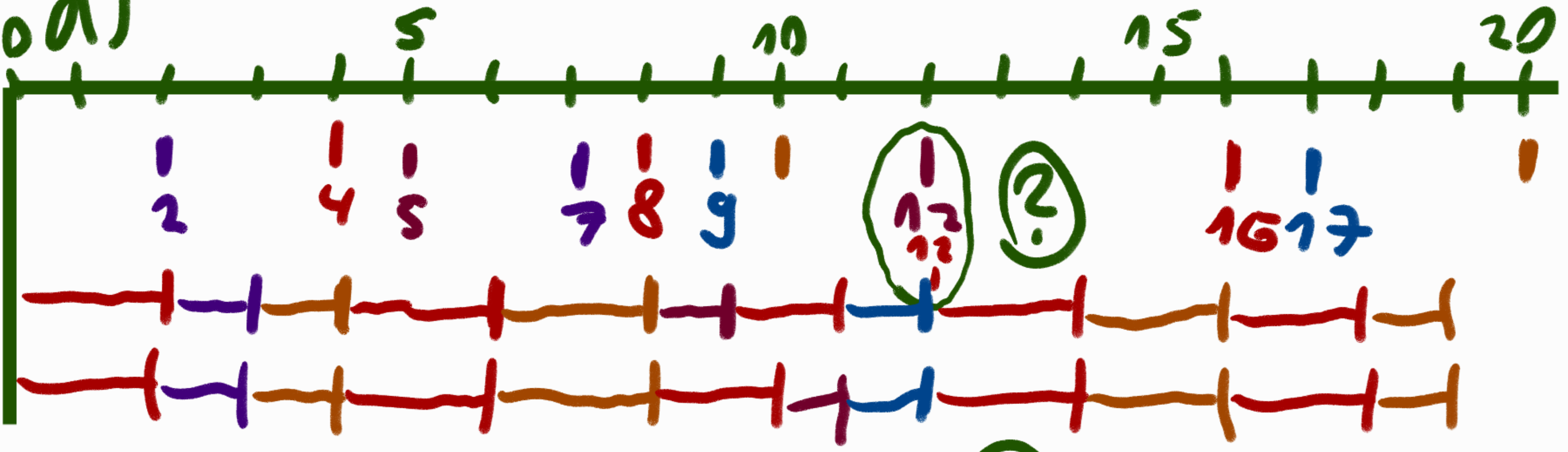
$$d_k = \max(r_k, d_{k-1}) + \frac{c_k}{u_s}$$

$$d_{r_{II}} = \max(7, 5) + 5 = 12$$

$$d_{r_{III}} = \max(12, 9) + 5 = 17$$

d)

$lcm(10,4) = 20$



→ RM ?

Exercise 4

We have to design a system that schedules periodic tasks with EDF and employs a total bandwidth server to serve aperiodic requests. We know of that the first aperiodic request has a computation time $C_a = 2$ and a relative deadline $D_a = 7$. What is the maximum processor utilization available for periodic tasks if we want to use the results about TBS to guarantee that this aperiodic task completes within its deadline.

$$d_a = a_a + D_a$$

$$a_a + D_a = a_a + \frac{C_a}{U_s}$$

$$\Leftrightarrow \cancel{a_a} + 7 = \cancel{a_a} + \frac{2}{U_s}$$

$$\Leftrightarrow U_s = \frac{2}{7}$$

U_p maximal if $U_p = 1 - U_s$

$$= \frac{5}{7}$$

Exercise 2

The feasability of a set of tasks with relative deadlines less than or equal to periods can be guaranteed with EDF by verifying that

$$\forall L > 0 \quad \sum_{i=1}^{n} \left\lfloor \frac{L + T_i - D_i}{T_i} \right\rfloor C_i \leq L.$$

Prove this theorem *only* for relative deadlines equal to periods. Note, use the theorem of the lecture: A set of periodic tasks $\tau_1, ..., \tau_n$ (with $D_i = T_i$) is schedulable with EDF iff $\sum_{i=1}^{n} \frac{C_i}{T_i} \leq 1$. (1)

(1): $\sum_{i=1}^{n} \frac{C_i}{T_i} \leq 1$

(2):

if $D_i = T_i$: $\forall L>0 \; \sum_{i=1}^{n} \left\lfloor \frac{L + \overbrace{T_i - D_i}^{0}}{T_i} \right\rfloor = \sum_{i=1}^{n} \left\lfloor \frac{L}{T_i} \right\rfloor \leq L$

Zz: (1) $\Leftrightarrow$ (2)

$\Rightarrow$: (1) $\Rightarrow$ (2)

If $U \leq 1$

$\forall L > 0: L \geq UL = \sum_{i=1}^{n} \left(\frac{C_i}{T_i}\right) \cdot L$

$= \sum_{i=1}^{n} \left(\frac{L}{T_i}\right) \cdot C_i$

Trans.

$$\geq \sum_{i=1}^{n} \left\lfloor \frac{L}{T_i} \right\rfloor \cdot C_i$$

$\Leftarrow$: (2) $\Rightarrow$ (1) $\equiv \neg C_1 \to \neg C_2$

If $U > 1$, then

$$\exists\, L > 0 : L < UL = \sum_{i=1}^{n} \left(\frac{C_i}{T_i}\right) \cdot L$$

$$= \sum_{i=1}^{n} \left(\frac{L}{T_i}\right) \cdot C_i$$

$$= \sum_{i=1}^{n} \left\lfloor \frac{L}{T_i} \right\rfloor \cdot C_i$$

for $L = \max(T_1, \ldots, T_n)$